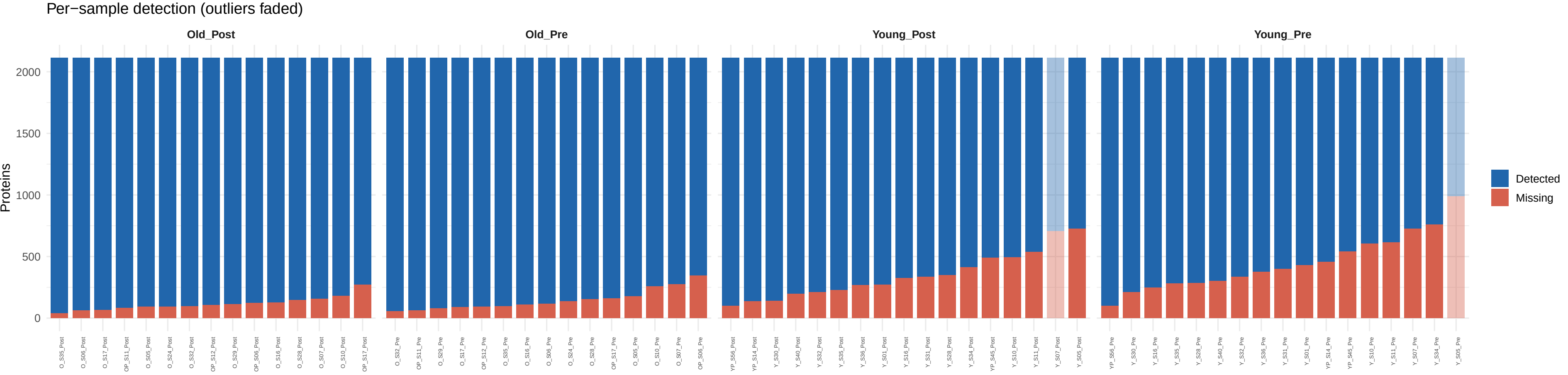
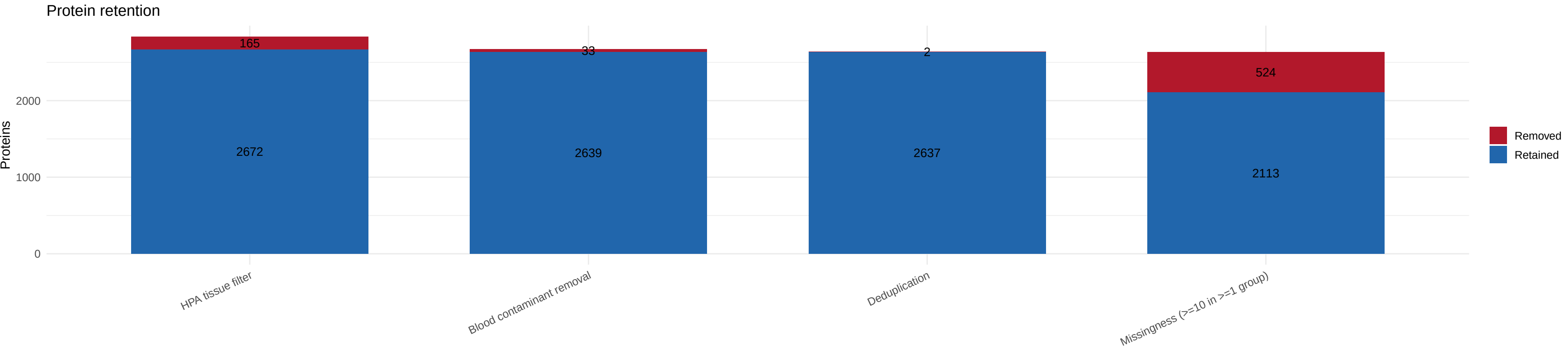


Protein Filtering & Detection

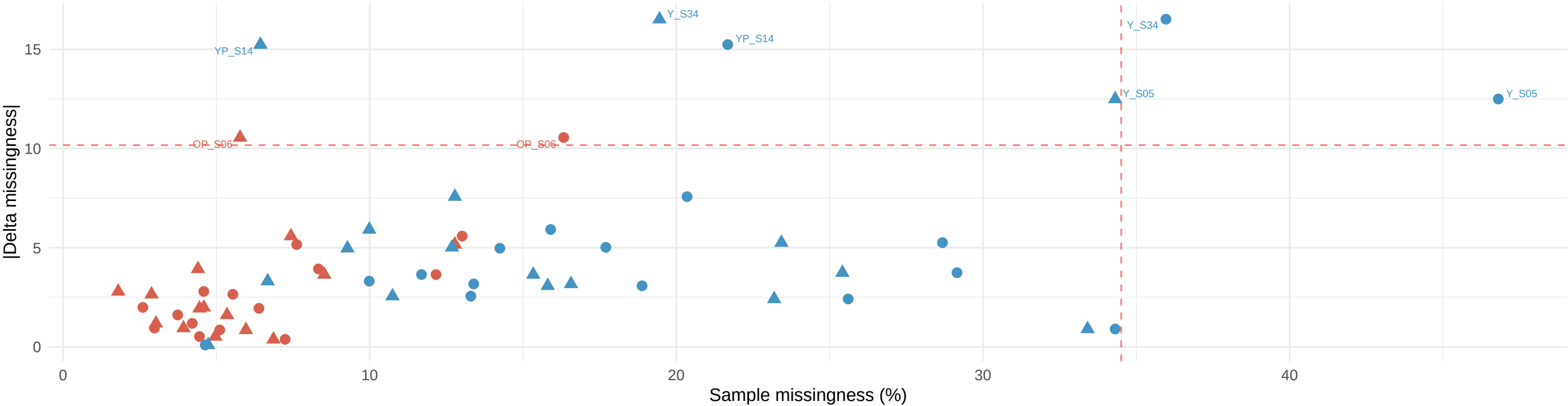
2837 raw -> 2113 retained | 62 samples



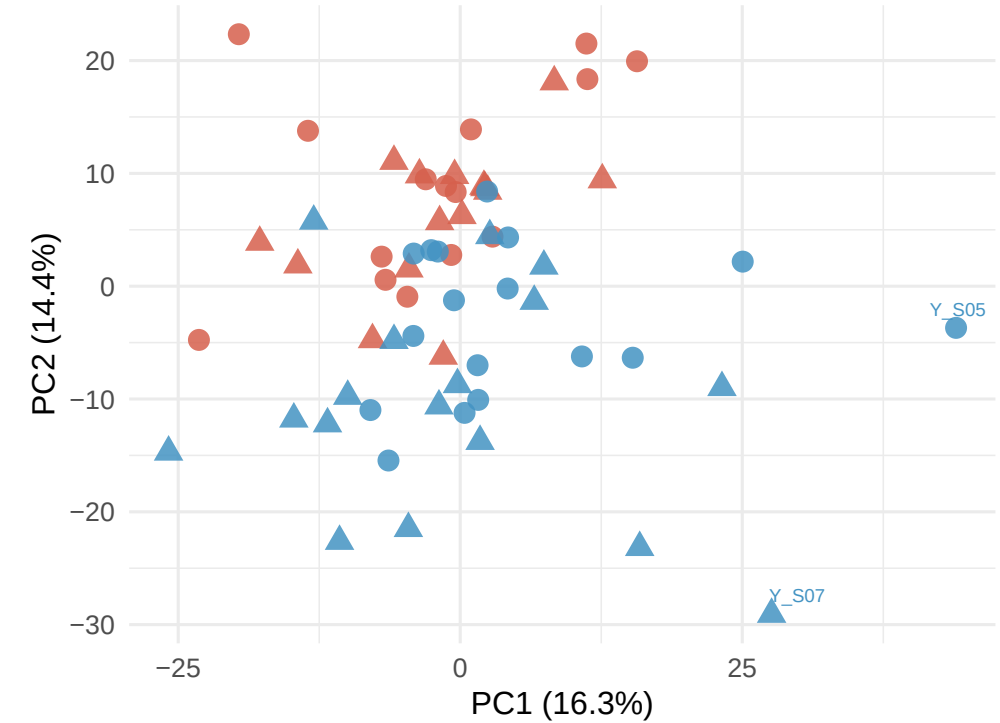
# Outlier Diagnostics (4-method consensus)

>=3/4 agreement | 2 removed

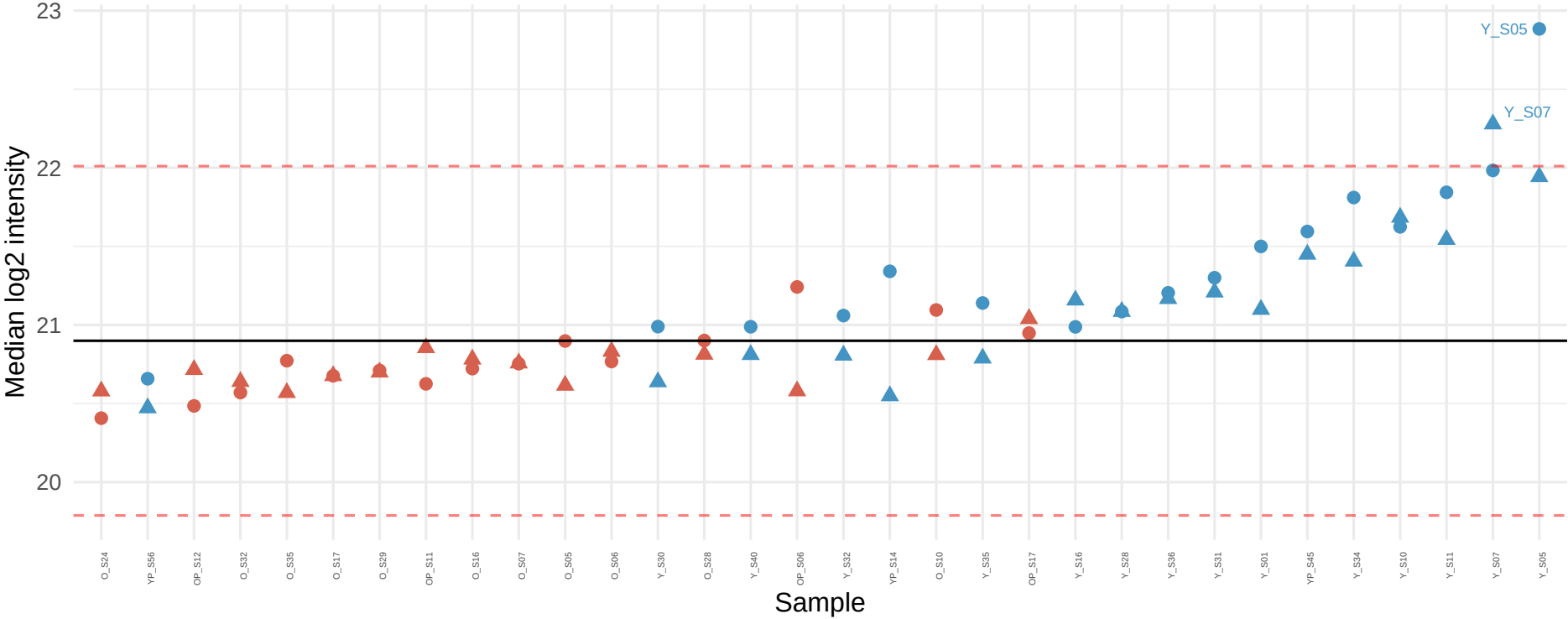
A: Missingness  
IQR thresholds: 34.5% / 10.2% | 8 flagged



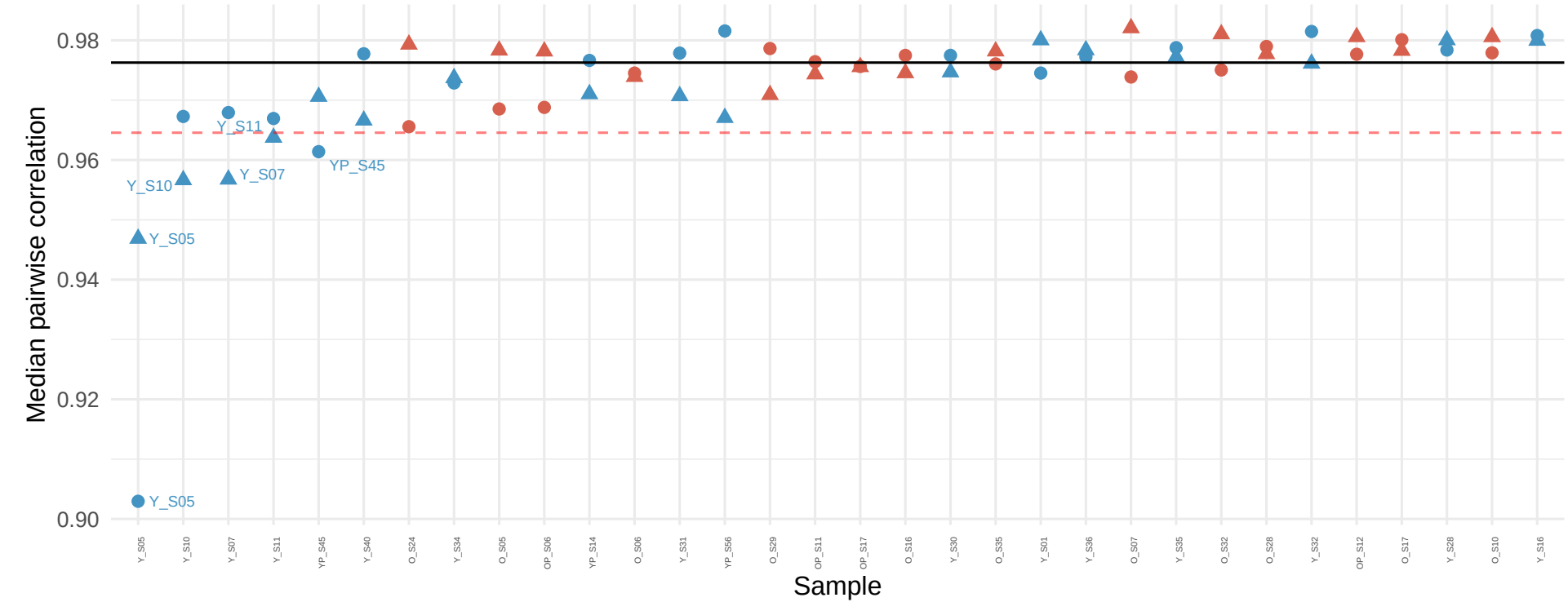
B: PCA Mahalanobis  
p < 0.01 | 2 flagged



C: MAD median intensity  
3x MAD | 2 flagged



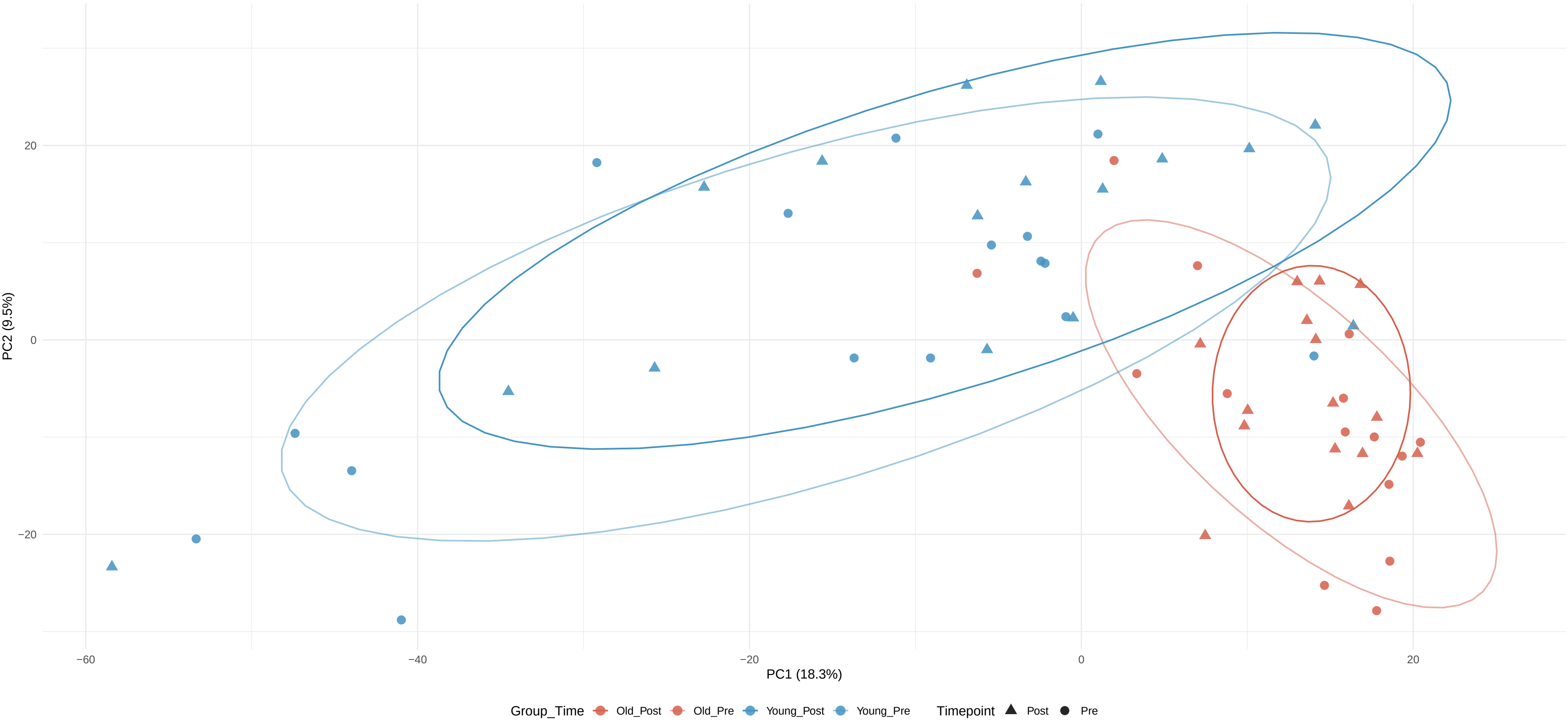
D: Inter-sample correlation  
3x MAD | 6 flagged



Timepoint ▲ Post ● Pre age ● Old ● Young

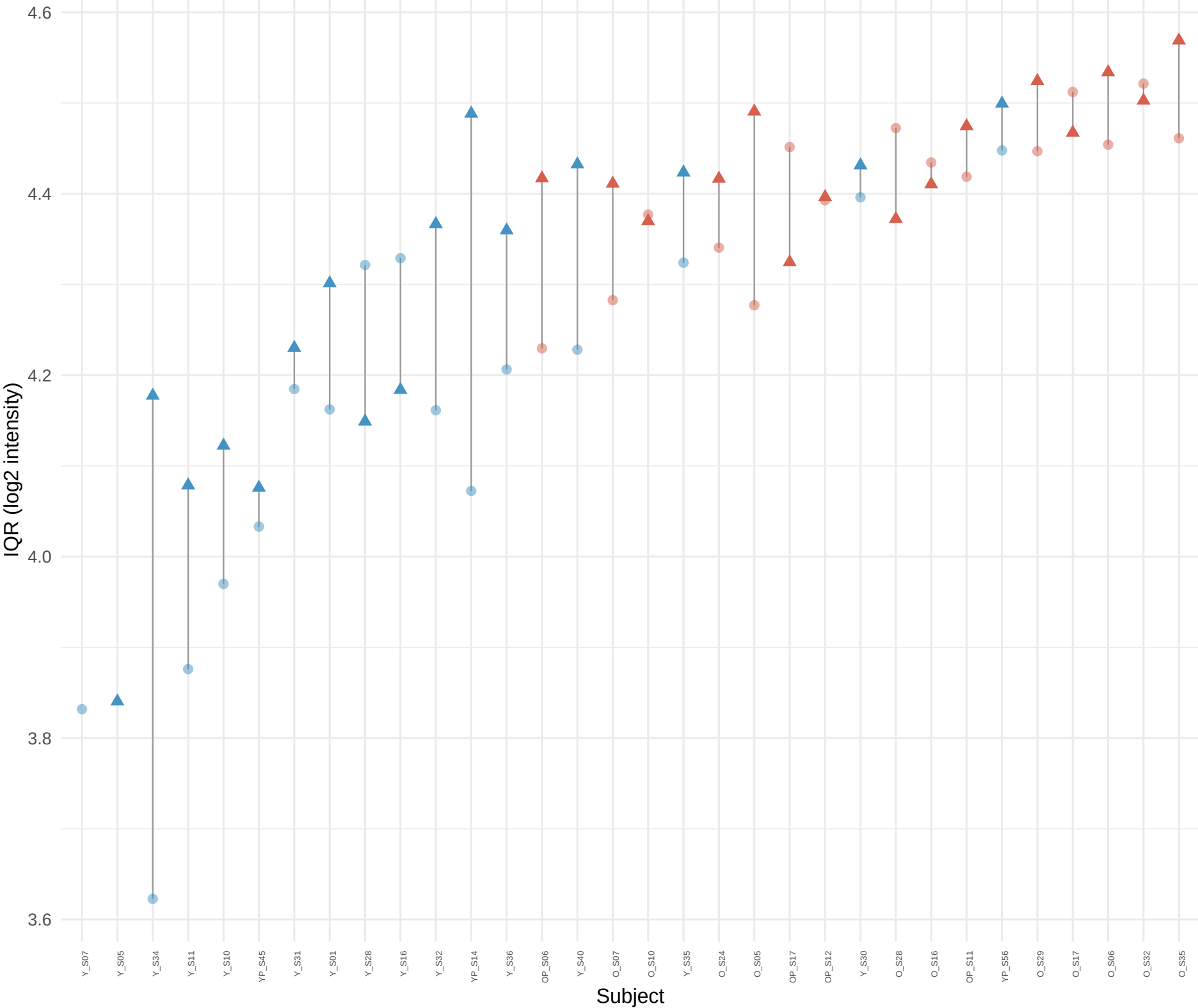
Post-Normalization QC

Post-normalization PCA

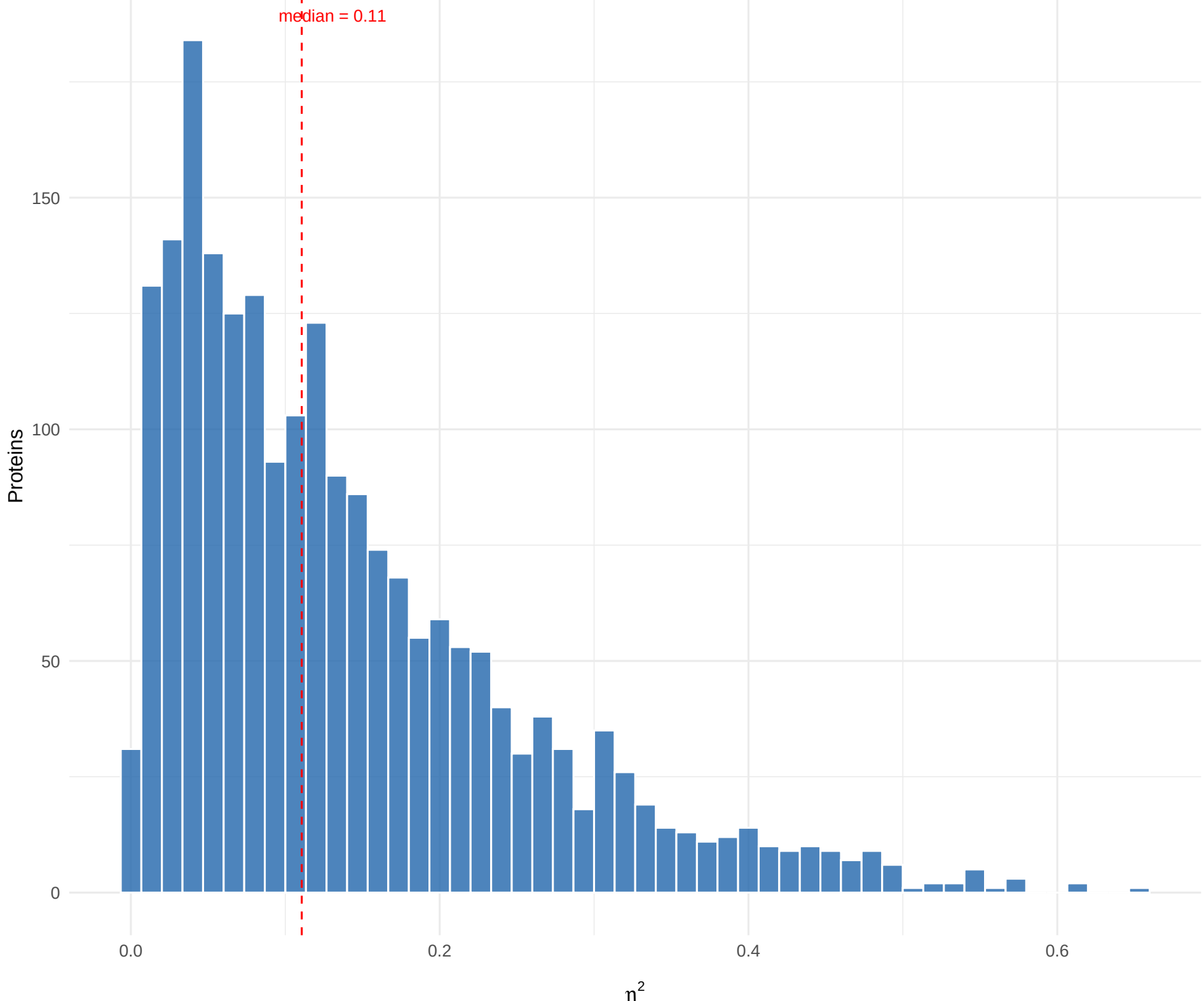


# Variability Summary

Per-subject variability  
32 subjects | lines connect Pre-Post pairs



Variance partition by group



Group\_Time ● Old\_Post ● Old\_Pre ● Young\_Post ● Young\_Pre Timepoint ▲ Post ● Pre